VORTEXA 3.0

12-HOUR HACKATHON · POWERED BY A.I.R.S · DR. D. Y. PATIL INSTITUTE OF TECHNOLOGY, PIMPRI

DOMAIN & PS NO.

6 · Public Systems & Civic Technology · P-03

PROBLEM STATEMENT

RailFlow: Railway Disruption & Platform Conflict Simulator

TEAM NAME

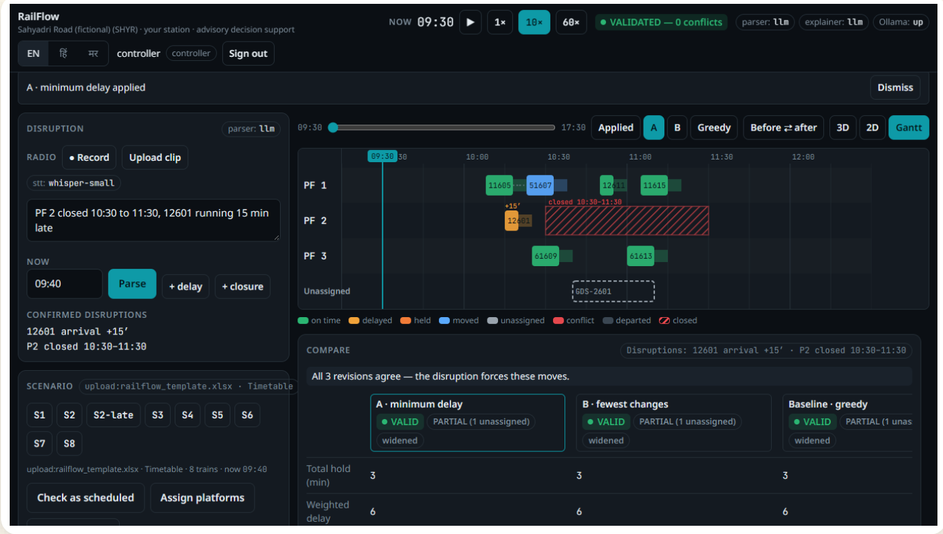
Runtime Renegades

TEAM ID

VTX26-92646

TEAM

Gurjas Singh Gandhi (Leader) · Aaditi Choramle
MCA, P.E.S. Modern College of Engineering, Pune



The planner may be wrong. The validator cannot be fooled.
The AI never touches a platform.

PROBLEM STATEMENT

At an Indian junction, one late train rarely stays one late train — and it is replanned by hand.

1 One delay becomes many

A train running 30 min late holds its platform; the trains due there next are blocked and the delay spreads down the timetable.

2 Replanned by phone and memory

Station masters and section controllers rework platforms over radio and phone, with registers, while the clock keeps running.

3 Too many rules to juggle

Minimum clearance between trains, 24–26-coach rakes vs short platforms, MEMU-only bays, goods loops, maintenance blocks.

4 A missed clash reaches the platform

Manual checks miss a 2-minute gap; late platform switches send crowds across foot-over-bridges.

WHAT P-03 ASKS FOR — AND WHAT RAILFLOW DELIVERS, ALL FIVE

✓ ASSIGN

No platform double-booked; every gap kept

✓ DETECT

Each conflict reported by train & platform

✓ REPLAN

Delays & closures, with a reason per change

✓ VALIDATE

Independent check; never 'valid' when not

✓ AI

Text → delays & closures; explains trade-offs; never assigns

PROPOSED SOLUTION: RAILFLOW

Speak or type the disruption. Get a replan that is proven conflict-free — in seconds.

1 REPORT

Radio clip, typing or the + delay / + closure forms · EN · हिं · मर

2 AI PROPOSES

Local LLM turns words into delays & closures, each linked to its source words

3 HUMAN CONFIRMS

Unknown trains are rejected, never guessed; the controller signs off

4 OPTIMISE

OR-Tools CP-SAT: A = least delay, B = fewest platform changes, plus a greedy baseline

5 PROVE & APPLY

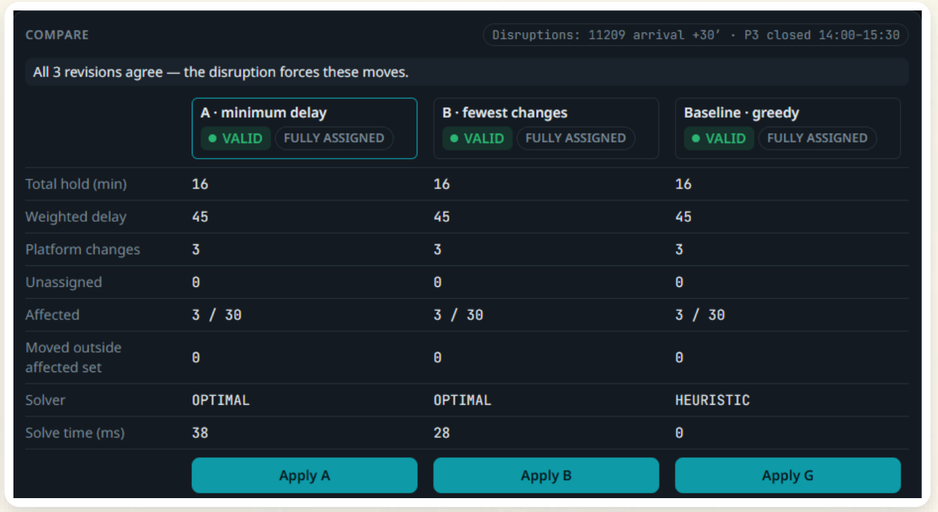
Independent validator (V1–V10) → hash-checked apply → audit chain

OUR DESIGN RULE

The planner may be wrong.
The validator cannot be fooled.
The AI never touches a platform.

BUILT FOR

- Station master / Dy. SS
- Section controller
- Divisional control
- Admin / audit



Live, S2: 11209 runs 30 min late + PF 3 shut → 3 of 30 trains move; all 3 revisions independently VALID.

TECHNICAL APPROACH

OPERATOR UI · React 19 + TypeScript

Gantt truth view · 3D / 2D station · compare & apply · reason cards · EN · हिं · मर · phone / tablet / desktop

API · FastAPI + Pydantic v2

Roles & Argon2id sessions · CSP + Origin checks · every action and refusal written to the audit chain

AI BRANCH (advisory)

Ollama qwen2.5:3b, local
Whisper small for radio
Regex + template fallback
Numbers must appear in the source

PLANNER

OR-Tools CP-SAT
Lexicographic objective
Replans the affected set only
Greedy baseline to compare

VALIDATOR (independent)

Sweep line, V1–V10
Imports only the data models
Reports every violating pair
Sole source of VALID

proposals → human confirm → plan → prove → apply (AI output can never express a platform assignment)

STORAGE · SQLite (WAL)

Immutable revisions · SHA-256 hash-chained audit log · apply = hash match + re-validate + one transaction

HOW IT DECIDES

One occupancy rule

A train holds its platform for [arrival, departure + min gap).

Strict priorities

First: no train left unplaced. Then the controller's goal: least delay or fewest changes.

Touch only what broke

Replan the affected trains; widen the set only when forced.

Nothing unproven ships

A plan that fails validation cannot be applied — the button is locked.

Python 3.12 · FastAPI · OR-Tools · SQLite · React 19 · Three.js · Ollama · faster-whisper · pytest + Hypothesis

HOW THE VALIDATOR WORKS

A separate checker re-proves every plan from scratch — with a different algorithm from the planner.

THE SWEEP LINE, ON THE REAL S3 CONFLICTS



Same answers as the station's own Excel answer key — found automatically.

TEN RULES, CHECKED ON EVERY PLAN

V1	Every timetable train exactly once	V6	Hold within the 45-min limit
V2	The platform exists	V7	Departed or berthed trains stay put
V3	Required / allowed platform kept	V8	No overlap, clearance kept (sweep)
V4	Stay = dwell + departure delay	V9	No train inside a closure
V5	Never earlier than the delay allows	V10	Coaches / wagons fit the platform

THE ALGORITHM, PER PLATFORM

1. Turn every assigned train into an interval [arr, dep + gap).
2. Sort by arrival — O(n log n).
3. Sweep: compare each train with the ones after it; stop at the first arrival ≥ dep + gap. Every clashing pair is reported, not just neighbours.
4. Measure in minutes: stay = min(dep A, dep B) − arr B; reserved = min(dep A, dep B) + gap − arr B; shortfall = dep A + gap − arr B.

WHY IT CANNOT BE FOOLED

- Imports only the data models and Python's standard library — a test fails the build if that changes.
- Different algorithm from the planner: a sweep line, not CP-SAT constraints.
- The only source of the VALID badge; re-run on the hashed plan at Apply.
- Proven: 400 / 400 random plans checked; every injected planner fault caught (S5).

FEASIBILITY AND VIABILITY

137

automated backend tests, all passing

400 / 400

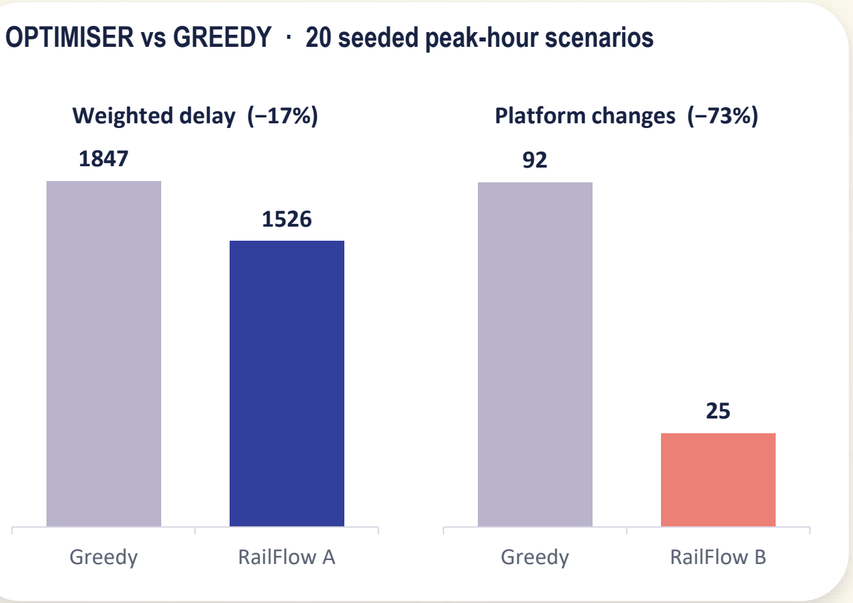
random disruption cases — every plan proven valid

9 / 9

wrong AI readings caught before reaching 'ready'

₹0

cloud cost · runs offline on one laptop, no GPU



Deployable today

One laptop, fully offline. Judges' and staff phones join over the hotspot — no cloud, no GPU.

Fits how stations work

Imports the working-timetable Excel; any station via template, JSON or typed in by hand.

Honest next steps

Route conflicts in the station throat (designed, measured), past-midnight trains, multi-station.

AI parser accuracy: LLM 44/52 exact vs regex 36/52 on labelled messages; when Ollama is off, RailFlow falls back to regex + templates and still plans.

IMPACT AND BENEFITS

Controllers

From minutes of phone calls to a checked replan in seconds — every move arrives with its reason, ready to announce.

Passengers

Option B keeps platform changes to a minimum: fewer last-minute dashes across foot-over-bridges.

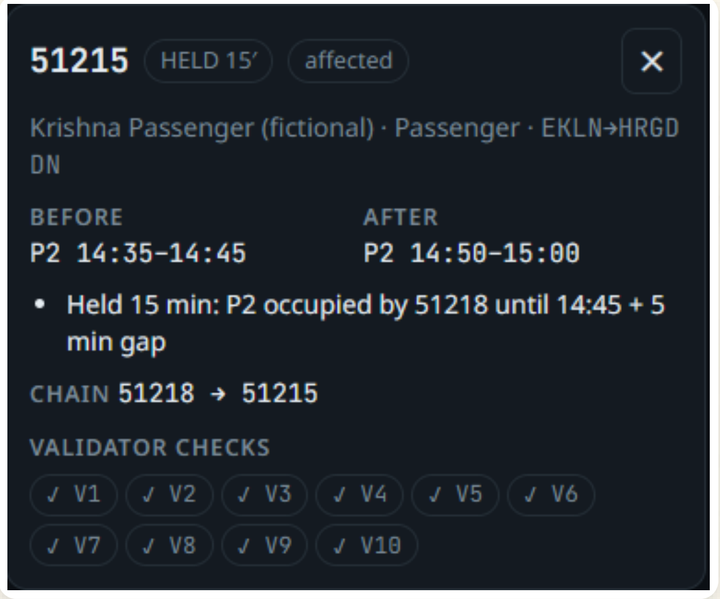
Railway management

Every confirm, plan, apply and refusal sits in a tamper-evident audit chain; admins audit, controllers act.

Safety & trust

No plan reaches a platform unless an independent validator proved it conflict-free.

Beyond the demo: any station via Excel / JSON template or manual entry · 3 languages · works on phone, tablet and desktop.



Every change explains itself: cause, chain and checks.

S8 · THE CONTROLLER CHOOSES, INFORMED

A
2 moves
weighted delay 76

B
1 move
weighted delay 136

DIFFERENTIATOR

	Phone + register	Optimiser / chatbot	RailFlow
Finds every clash, with minutes	No	Partly	Yes
Moves only the affected trains	Partly	No	Yes
Independent proof each plan is valid	No	No	Yes
AI can never assign a platform	—	No	Yes
Explains why each train moved	Partly	Partly	Yes
Voice · EN · हिं · मर · offline	Yes	No	Yes
Tamper-evident audit trail	No	No	Yes



Our boldest demo: **we break our own planner, live.**

The validator catches the overlap — NOT VALID, Apply locked. Trust is proven, not promised.

We don't ask anyone to trust the AI or the optimiser. Every plan is proven before it touches a platform — and the AI only listens and explains.

Team Runtime Renegades · VTX26-92646 · P-03 RailFlow

github.com/Gurjas2112/RailFlow

THANK YOU!

VORTEXA 3.0 · AIRS

Game's Still On

